Preliminary GEK-45162

**GE Drive Systems**

**INSTRUCTIONS**

# BCD ASSEMBLY

# 7V11CBD Option Kit

## IMPORTANT INFORMATION

**CAUTION:** To ensure proper operation of the QIII DC Controller, these instructions must be followed for proper installation and set-up of this Option Kit.

*These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.*

# SAFETY PRECAUTIONS

## WARNING - SHOCK AND BURN HAZARD:

**Always disconnect power to the drive before performing any work on drive components. Failure to do so may cause serious injury to personnel and damage to the drive or the driven machinery.**

**NOTE:** *Read all WARNINGS, CAUTIONS and NOTES in the QIII DC Controller Instruction Book (GEK-45167) prior to installing this Option Kit.*

**NOTE:** *Read and heed all WARNING, CAUTION, and NOTE labels posted on the equipment prior to installing this Option Kit.*

# KIT DESCRIPTION AND CONTENTS

**NOTE:** *Application Card Option Kit (7V11CAC) must be installed prior to installing this kit.*

## DESCRIPTION:

This optional kit provides an interface to the drive through a terminal block and ribbon cable connector to the Application card. The terminal block has individual terminals for discrete wiring connections that are prewired to pins on its ribbon cable receptacle. The terminal block can be mounted at any customer selected mounting point within **10 feet** of the controller.

## CONTENTS:

The kit contains the following components (in addition to these instructions):

| Component | Part Number | Qty. |
|---|---|---|
| Terminal Block (with mounting rail) | 104X154AB038 | 1 |
| End Caps | 104X154AB039 | 2 |
| 16PL Connector Cable (10' length) | 36A358028ARG01 | 1 |

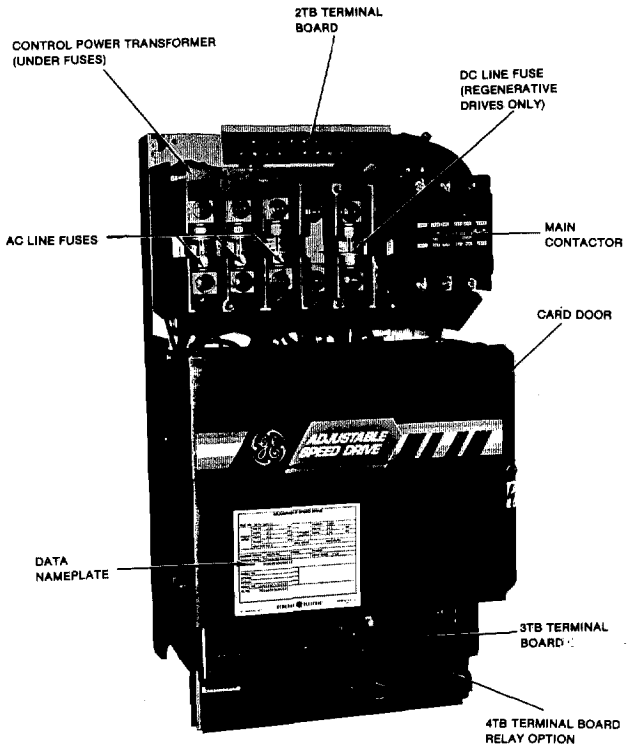



**Figure 1.** QIII DC CONTROLLER

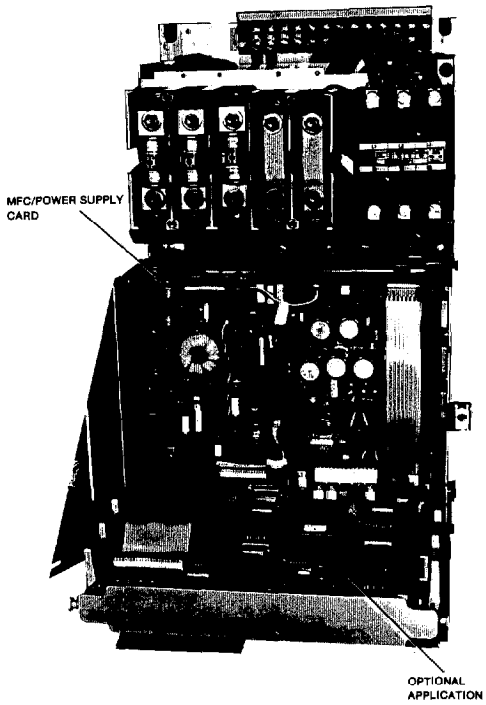



**Figure 2.** CONTROLLER WITH FRONT CARD RACK LOWERED

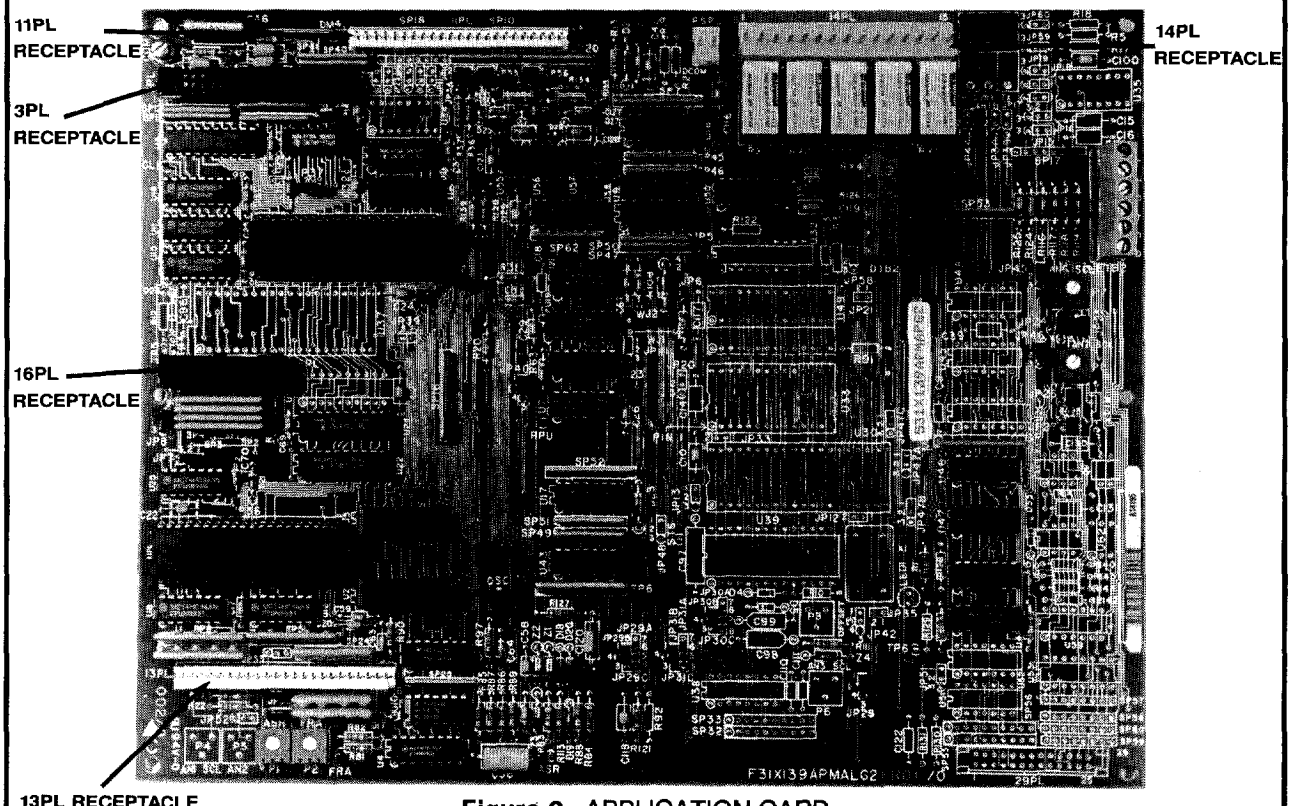



**Figure 3.** APPLICATION CARD

# KIT INSTALLATION PROCEDURE

**WARNING - SHOCK AND BURN HAZARD:**
**Always disconnect power to the drive before removing or inserting a printed circuit card. Failure to do so may cause serious injury to personnel and damage to the drive or the driven machinery.**

1. Verify that power to the controller that the option kit is to be installed is OFF.

2. Loosen thumbscrew (located on right side) and open the controller card door (see Figure 1).

3. Access the Application card as follows (see Figure 2):
   a. Slide each card rack latch bar forward (located on each side of controller).
   b. Lift front card rack upward and allow to tilt forward until it rests in horizontal position.

4. Disconnect 13PL ribbon cable from its receptacle to access 16PL receptacle by pressing holding tabs back while pulling on sides of connector plug until free.

5. Install 16PL as follows (see Figure 3):
   a. Orient 16PL ribbon cable to its receptacle on the Application card with cable extending toward bottom of card rack and press the connector into fully seated position.
   b. Route ribbon cable under bottom of the card rack.
   c. **For "G" and "C" frames only,** route 16PL ribbon cable behind 3TB Terminal Board card.

6. Reconnect 13PL ribbon cable to its receptacle by pressing connector plug in until holding tabs snap into place.

7. Raise the card rack upward to a vertical position and slide rack extensions into slots provided on top of controller until card rack is fully seated in position.

8. Slide the card rack latch bars (located on each side of controller) back.

9. Close the controller card rack door and tighten the thumbscrew (located on right side, see Figure 1).

10. Route 16PL ribbon cable (from behind 3TB on "G" and "C" frames or from below card rack enclosure on "D" frames) to the mounting point of the terminal block.

**CAUTION:** Ensure that there is enough free cable at the card rack end to allow raising and lowering of the card rack and that 16PL ribbon cable is adaquetaly secured to prevent any interference or abrasion before permanently mounting terminal block.

11. Secure mounting rail in selected location with appropriate mounting hardware (mounting hardware not provided).

12. Slide terminal block onto mounting rail and center it on the mounting rail.

13. Slide an end cap onto each end of the mounting rail and position against the ends of the terminal block, then tighten the two screws in each end cap to secure the terminal block in place.

14. Make sure 16PL ribbon cable is not twisted and press plug connector into receptacle on terminal block until fully seated.

15. Make desired wiring connections to respective terminal connections of terminal block (see Table 1). (REF24 and PCOMX are used for power supply.)

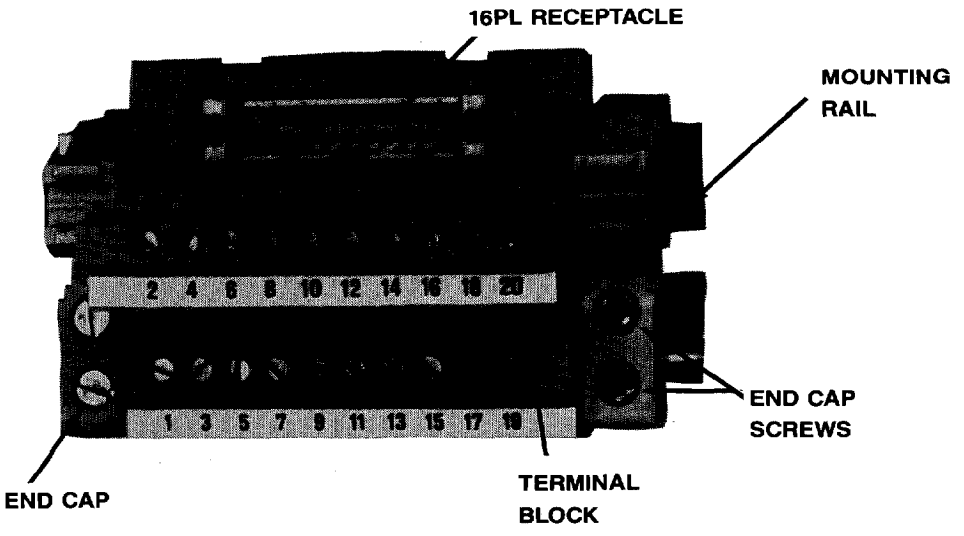



**Figure 4.** TERMINAL BLOCK WITH 16PL RECEPTACLE

## TABLE 1: TERMINAL BLOCK/16PL CONNECTOR POINT DESCRIPTIONS

| 16PL Pin | Terminal Number | Nomen-clature | Description |
|---|---|---|---|
| 1 | 1 | ---- | Not Used |
| 2 | 2 | ---- | Not Used |
| 3 | 3 | PI0 | Optional parallel word input, least significant binary bit input. (For inputs refer to GEK-45167 Table 1, Application card JP8 description for voltage level select.) Input must be set for 50 microseconds prior to strobe. |
| 4 | 4 | PI1 | Optional parallel binary bit input 1. |
| 5 | 5 | PI2 | Optional parallel binary bit input 2. |
| 6 | 6 | PI3 | Optional parallel binary bit input 3 (or PI0 - PI3 = BCD units digit). |
| 7 | 7 | PI4 | Optional parallel binary bit input 4. |
| 8 | 8 | PI5 | Optional parallel binary bit input 5. |
| 9 | 9 | PI6 | Optional parallel binary bit input 6. |
| 10 | 10 | PI7 | Optional parallel binary bit input 7 (or PI4 - PI7 = BCD tens digit). |
| 11 | 11 | PI8 | Optional parallel binary bit input 8. |
| 12 | 12 | PI9 | Optional parallel binary bit input 9. |
| 13 | 13 | PI10 | Optional parallel binary bit input 10. |
| 14 | 14 | PI11 | Optional parallel binary bit input 11 (or PI8 - PI11 = BCD hundreds digit). |
| 15 | 15 | PI12 | Optional parallel binary bit input 12. |
| 16 | 16 | PI13 | Optional parallel binary bit input 13. |
| 17 | 17 | PI14 | Optional parallel binary bit input 14 (or PI12 - PI14 = BCD thousands digit). |
| 18 | 18 | PI15 | Optional parallel sign bit input. |
| 19 | 19 | ---- | Not Used |
| 20 | 20 | ---- | Not Used |

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